

TURA, Russian Federation

WMO: 245070

Lat: 64.271N

Lon: 100.217E

Elev: 191

StdP: 99.05

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-50.1	-48.1	-51.6	0.0	-49.9	-49.7	0.0	-48.1	5.6	-22.6	4.8	-22.8	0.4	310	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	28.4	17.2	26.2	16.3	24.0	15.2	18.4	25.7	17.3	23.8	16.3	22.2	2.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.5	20.2	14.9	10.8	19.3	13.7	10.0	18.9	52.9	25.7	49.6	23.8	46.5	22.4	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.1	4.3	DB	-50.6	31.3	2.4	1.7	-52.3	32.5	-53.7	33.5	-55.0	34.5	-56.7	35.7	(4)
(5)			WB	-50.8	20.0	2.5	1.0	-52.6	20.7	-54.0	21.3	-55.4	21.8	-57.2	22.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.9	-34.3	-29.5	-16.8	-5.6	4.0	14.4	16.9	12.9	5.3	-6.3	-24.3	-33.0	(6)
(7)		DBStd	19.66	8.86	8.87	9.03	6.46	4.95	4.55	3.58	3.59	4.13	7.27	10.74	9.91	(7)
(8)		HDD10.0	7017	1372	1106	831	469	196	13	1	14	149	504	1028	1334	(8)
(9)		HDD18.3	9624	1630	1339	1089	719	445	134	72	172	390	763	1278	1593	(9)
(10)		CDD10.0	478	0	0	0	0	9	143	213	103	9	0	0	0	(10)
(11)		CDD18.3	43	0	0	0	0	0	14	26	3	0	0	0	0	(11)
(12)		CDH23.3	591	0	0	0	0	5	229	315	41	1	0	0	0	(12)
(13)		CDH26.7	149	0	0	0	0	0	68	77	4	0	0	0	0	(13)
(14)	Wind	WSAvg	1.6	1.0	1.1	1.6	2.2	2.3	2.0	1.9	1.7	1.8	1.3	1.0	(14)	
(15)	Precipitation	PrecAvg	366	15	12	11	16	36	46	52	71	36	32	23	17	(15)
(16)		PrecMax	545	27	23	24	29	81	104	122	141	80	57	47	45	(16)
(17)		PrecMin	243	5	1	3	5	6	9	7	9	3	4	8	2	(17)
(18)		PrecStd	78	5	5	6	7	18	21	33	35	19	14	11	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.3	-5.3	5.3	11.5	23.8	31.9	31.2	27.0	20.1	8.6	2.1	-4.7	(19)
MCWB			-11.3	-6.6	1.2	5.2	12.3	17.4	19.0	17.2	12.6	5.5	-0.1	-5.6	(20)	
2%		DB	-13.8	-8.6	1.7	8.2	19.1	28.6	28.8	24.3	16.9	6.0	-0.7	-9.2	(21)	
		MCWB	-14.4	-9.5	-1.1	3.1	10.3	16.7	17.6	15.7	11.2	3.6	-2.0	-9.7	(22)	
5%		DB	-17.3	-11.7	-0.9	5.8	15.2	25.9	26.9	22.0	14.2	4.0	-3.7	-13.7	(23)	
		MCWB	-17.8	-12.5	-3.2	2.1	7.9	15.7	16.9	14.6	9.9	2.3	-4.8	-14.2	(24)	
10%		DB	-20.7	-15.8	-3.2	3.9	12.4	23.5	24.9	19.7	12.0	2.2	-8.2	-17.6	(25)	
		MCWB	-21.0	-16.3	-5.0	0.7	6.6	14.2	16.0	13.9	8.7	0.6	-9.1	-18.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.2	-6.6	1.4	5.9	13.1	18.8	20.1	18.9	14.0	5.9	0.4	-5.5	(27)
MCDWB			-10.3	-5.6	4.7	10.1	22.3	27.9	28.5	25.0	17.4	8.1	1.9	-4.8	(28)	
2%		WB	-14.4	-9.6	-1.1	3.8	10.8	17.5	18.8	16.9	12.1	4.0	-1.8	-9.8	(29)	
		MCDWB	-13.9	-8.6	1.5	7.3	17.5	26.2	26.7	21.8	15.0	5.4	-0.6	-9.2	(30)	
5%		WB	-17.8	-12.5	-3.0	2.4	8.6	16.4	17.9	15.6	10.7	2.4	-4.9	-14.2	(31)	
		MCDWB	-17.4	-11.7	-0.9	5.3	14.2	24.3	24.4	19.9	13.7	3.9	-3.7	-13.7	(32)	
10%		WB	-21.0	-16.3	-5.0	1.1	6.8	15.3	17.0	14.6	9.1	0.8	-9.0	-18.0	(33)	
		MCDWB	-20.8	-15.8	-3.1	3.6	12.0	21.9	22.6	18.4	11.8	2.0	-8.1	-17.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.7	8.6	13.6	11.8	9.4	11.5	11.0	9.4	8.0	6.4	6.9	6.9	(35)
MCDBR			9.3	9.7	12.6	11.2	14.7	16.1	15.5	14.7	12.9	6.5	7.5	11.0	(36)	
MCWBR			9.0	8.9	10.2	7.4	7.8	6.3	6.1	6.8	7.4	4.9	7.0	10.5	(37)	
5% WB		MCDBR	9.4	9.6	12.4	10.1	13.1	14.1	12.7	10.9	11.0	6.3	7.7	10.8	(38)	
		MCWBR	9.1	9.0	10.3	7.1	7.5	6.0	5.6	6.0	6.9	4.9	7.2	10.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.212	0.265	0.299	0.333	0.347	0.369	0.437	0.413	0.330	0.272	0.195	0.199	(40)	
taud		1.876	2.021	2.062	2.067	2.235	2.420	2.199	2.228	2.436	2.223	1.900	1.870	(41)		
Ebn at Noon		314	621	768	828	858	848	763	743	756	625	332	112	(42)		
Edh at Noon		29	67	100	126	118	101	123	109	71	54	26	11	(43)		
(44)	All-Sky Solar Radiation	RadAvg	0.17	0.96	2.47	4.33	5.31	5.62	4.95	3.35	2.05	0.89	0.28	0.06	(44)	
(45)		RadStd	0.02	0.07	0.19	0.28	0.39	0.58	0.46	0.27	0.26	0.11	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.04	N/A	N/A	+1.76	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		+1.04	N/A	N/A	+1.76	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page